M. Naqib Rahimi, Ph.D.

Naqib.Rahimi@stonybrook.edu Naqib.Rahimy123@gmail.com Naqib Rahimi M. Naqib Rahimi
M. Naqib Rahimi http://naqibr.github.io/ +1-631-384-4156

Summary:
I am a Ph.D. candidate in Civil Engineering specializing in fluid-structure interaction, failure modeling, and finite element analysis. I have extensive experience developing advanced algorithms for failure simulations in multi-physics environments, with a focus on Smoothed Particle Hydrodynamics (SPH), Peridynamics, and Phase Field methods. I also possess hands-on expertise in high-performance computing, including CUDA parallelization.

EDUCATION

2021 – Now **Ph.D. Civil engineering., Stony Brook University**
Thesis title: Computational mechanics of extreme events: Advanced multi-physics simulations with Smoothed Particle Hydrodynamics, Isogeometric Analysis, Micro Plane Model, and Phase Field.

2019 – 2021 **M.Sc. Material Science and Nanoengineering, Sabanci University**
Thesis title: Peridynamic Modeling of Internal Features and Interfaces for Material Toughening.

2015 – 2021 **B.Sc. Mechanical Engineering, Inonu University**
Thesis title: Energy Assessment of Bio-wastes in Afghanistan's Kunar city.

2014 – 2018 **B.Sc. Civil Engineering, Inonu University**. Honors degree
Thesis title: Excel-based Programming in Civil Engineering.

JOB EXPERIENCE

May 2024 – Aug 2024 **Software Developer Intern,** ANSYS-LSDYNA.

- Continued working on similar topics from the previous year.
- Took on the integration of CUDA parallelization into parts of the ANSYS-LSDYNA code.
- Led the first steps in transforming a significant portion of the ANSYS-LSDYNA base code to CUDA parallel architecture.

May 2023 – Aug 2023 **Software Developer Intern,** ANSYS-LSDYNA.

- Developed parallel algorithms for ANSYS-LSDYNA software to simulate airbag behavior during car crashes.
- Conducted FEM analyses and meshing for complex geometries.
- Performed mesh cleaning and re-meshing using ANSA and LS-PrePost.

Jun 2017 – Aug 2017 **Engineer Intern,** Inonu University Department of Constructions.

- Led a team of 9 interns in construction activities at Inonu University's campus.
- Participated in the construction of various new buildings, including a bridge, amphitheater, and student activity center.

Jun 2016 – Aug 2016 **Engineer Intern,** Guven Constructions.

- Reported workflow and material usage for ongoing construction sites.
- Engaged in regular standardization and quality control activities.

PROJECTS

2020 – 2022 **Parallelized Hybrid Particle Methods Supported by Innovative Non-local Models: Applications to Fluid-Structure-Interaction Analysis**
Research Assistant (Project ID: B.A.KM-21-02377)

- Developed a new SPH-Peridynamics numerical method for failure analysis of structures in multi-physics scenarios.
- Applied the method to complex cases and published the findings in various articles.

2019 – 2021 **Structural Health Monitoring of Sandwich Composites by Combining Peridynamics and iFEM Methodology: Theoretical Modeling, Numerical Analysis, and Experimental Verification**
Research Assistant (TUBITAK Project ID: 217M207)

- Performed various FEM analysis on composite structures.
- Developed numerical techniques for health monitoring of airplane wings.
- Created C++ algorithms to model damage propagation in composite materials.
- Assisted in the manufacturing and testing of a composite airplane wing.

2017 – 2018 **Design of Unmanned Ground Vehicle for Anti-Terrorism and Defense**
Research Assistant (BAP Project ID: FLO-2018-1028)

- Participated in designing a UGV for anti-terrorism purposes.
- Used FEM to model the vehicle and analyze its behavior under various conditions.

SKILLS

Coding	C++, Python, Fortran, MATLAB, CUDA, and FEniCS.
Methods	SPH, Phase Field, Finite Element, Isogeometric Analysis, Peridynamics.
Softwares	ANSYS, SolidWorks, AutoCAD.
Post/Pre-processing	ANSA, LS-PrePost, ParaView, GiD, and Corel Draw.
Languages	English, Persian, Turkish, Hindi, and Pashto.

AWARDS

2024 **IACS Junior Researcher Award**
Presented by the Institute for Advanced Computational Science (IACS) to PhD students for their outstanding research achievements.

SBU Civil Engineering Research Merit Award
Recognizes exceptional research contributions from a graduate student in the Department of Civil Engineering at Stony Brook University.

2023 **First Place Award**
Awarded first place by the Fluid Dynamics Committee of the U.S. Engineering Mechanics Institute (EMI), a technical committee of the American Society of Civil Engineers (ASCE).

2019 **Summa Cum Laude Award**
Ranked 1st out of 940 students in the Faculty of Engineering, Inonu University.

2018 **First Place in National Undergraduate Thesis Competition**
Awarded by the Scientific and Technological Research Council of Turkey (TUBITAK) for outstanding undergraduate thesis work.

2017 **Young Merit Award**
Presented by the MISA Organization for exceptional academic and personal achievements.

2011 **Silver Medalist in Mathematics**
Achieved silver at the International Science Olympiad (ISO) in Abuja, Nigeria.

2010 **Gold Medalist in Mathematics**
Awarded the gold medal at the National Mathematics Olympiad in Afghanistan.

SCHOLARSHIPS

2019 **Master's Scholarship**
2015 **Undergraduate Scholarship (Mechanical Engineering)**
2013 **Presidential Scholarship (Civil Engineering)**
2008 **High School Scholarship**

TEACHING EXPERIENCE

Jan 2024 – May 2024	**Steel and Reinforced Concrete Design I (CIV312)**
Jan 2021 – Jul 2021	**Calculus (Math102)**
Sep 2020 – Jan 2021	**Manufacturing Processes (IE309)**
Jan 2020 – June 2020	**Int. to Material Science (ENS205)**
Jan 2019 – Jan 2020	**Calculus (MATH102)**
Jan 2019 – Apr 2019	**Dynamics (DNK201)**

PUBLICATIONS

Journal Articles

[10] **M. N. Rahimi**, G. Moutsanidis, and L. Svolos, "Phase field modeling of dynamic brittle fracture in functionally graded materials under thermal shock," *Currently working on*, Nov. 2024.

[9] **M. N. Rahimi** and G. Moutsanidis, "Iga-sph: Coupling isogeometric analysis with smoothed particle hydrodynamics for air-blast–structure interaction," ***Engineering with Computers***, pp. 1–22, May 2024. DOI: https://doi.org/10.1007/s00366-024-01978-0.

[8] **M. N. Rahimi** and G. Moutsanidis, "An sph-based fsi framework for phase-field modeling of brittle fracture under extreme hydrodynamic events," ***Engineering with Computers***, Aug. 2023. DOI: https://doi.org/10.1007/s00366-023-01857-0.

[7] **M. N. Rahimi**, D. C. Kolukisa, M. Yildiz, M. Ozbulut, and A. Kefal, "A Generalized Hybrid Smoothed Particle Hydrodynamics-Peridynamics Algorithm with a Novel Lagrangian Mapping for Solution and Failure Analysis of Fluid-Structure Interaction Problems," ***Computer Methods in Applied Mechanics and Engineering***, Feb. 2022, (Highly cited). DOI: 10.1016/j.cma.2021.114370.

[6] **M. N. Rahimi** and G. Moutsanidis, "A smoothed particle hydrodynamics approach for phase field modeling of brittle fracture," ***Computer Methods in Applied Mechanics and Engineering***, Aug. 2022. DOI: https://doi.org/10.1016/j.cma.2022.115191.

[5] **M. N. Rahimi** and G. Moutsanidis, "Modeling dynamic brittle fracture in functionally graded materials using hyperbolic phase field and smoothed particle hydrodynamics," ***Computer Methods in Applied Mechanics and Engineering***, Nov. 2022. DOI: https://doi.org/10.1016/j.cma.2022.115642.

[4] **M. N. Rahimi**, A. Kefal, and M. Yildiz, "An improved ordinary-state based peridynamic formulation for modeling FGMs with sharp interface transitions," ***International Journal of Mechanical Sciences***, May 2021. DOI: 10.1016/j.ijmecsci.2021.106322.

[3] B. AlKhateab, I. E. Tabrizi, J. S. M. Zanjani, *et al.*, "Damage mechanisms in CFRP/HNT laminates under flexural and in-plane shear loadings using experimental and numerical methods," ***Composites Part A:*** *Applied Science and Manufacturing*, Oct. 2020. DOI: 10.1016/j.compositesa.2020.105962.

[2] **M. N. Rahimi**, A. Kefal, M. Yildiz, and E. Oterkus, "An ordinary state-based peridynamic model for toughness enhancement of brittle materials through drilling stop-holes," ***International Journal of Mechanical Sciences***, Sep. 2020. DOI: 10.1016/j.ijmecsci.2020.105773.

[1] **M. N. Rahimi** and O. H. Bettemir, "Development of an Unmanned Ground Vehicle for Shelter and Cave Reconnaissance and annihilation," *Savtek 2018 9. Defence Technologies*, vol. 1, no. -, pp. 761–771, Apr. 2018.

Conference Proceedings

[8] **M. N. Rahimi**, G. Moutsanidis, and L. Svolos, "Dynamic crack propagation in functionally graded materials under thermal shock: A novel phase field approach," in *Engineering Mechanics Institute Conference and Probabilistic Mechanics and Reliability Conference (EMI/PMC 2024) in Chicago, Illinois, May 28-31, 2024*, May 2024.

[7] **M. N. Rahimi** and G. Moutsanidis, "High Fidelity Modeling of Fracture Under Extreme Hydrodynamic Events: A Coupled SPH-Phase-Field FSI Approach," in *Engineering Mechanics Institute Conference 2023 (EMI 2023) in Atlanta, Georgia, June 6-9, 2023*, Jun. 2023.

[6] D. C. Kolukisa, R. Saghatchi, **M. N. Rahimi**, G. Moutsanidis, and M. Yildiz, "SPH-PD Modeling of the Periodic Elastic Response of a Beam Behind a Cylinder in Laminar Incompressible Flow," in *The 12th TSME International Conference on Mechanical Engineering in Phuket, Thailand*, Dec. 2022.

[5] G. Moutsanidis and **M. N. Rahimi**, "SPH framework for Modeling Fracture in Fluid-Structure Interaction: a comparative study between phase field and peridynamics," in *Meshfree and Novel Finite Element Methods with Applications in Berkeley, California*, Jun. 2022.

[4] **M. N. Rahimi** and G. Moutsanidis, "A coupled total Lagrangian SPH-phase field framework for brittle fracture," in *16th SPHERIC in Catania, Italy*, Jun. 2022.

[3] **M. N. Rahimi** and G. Moutsanidis, "SPH framework for Hyperbolic phase field modeling of brittle fracture," in *Meshfree and Novel Finite Element Methods with Applications in Berkeley, California*, Jun. 2022.

[2] **M. N. Rahimi**, A. Kefal, and M. Yildiz, "Numerical Investigation on Effective Toughening Mechanisms of Graded Composites," in *JOINT EVENT: ICCS23 - 23rd International Conference on Composite Structures & MECHCOMP6 - 6th International Conference on Mechanics of Composites*, Elsevier, Mar. 2020, p. 151.

[1] **M. N. Rahimi**, O. F. Bulak, and O. H. Bettemir, "Excel Based Program Modeling and Project Management Application Development.," in *Tubitak Project Competition Proceedings*, The Scientific and Technological Research Council of Turkey (TUBİTAK), 2018, p. 87.

References

Assist. Prof. Georgios Moutsanidis
Professor
Department of Civil Engineering
Institute for Advanced Computational Science
Stony Brook University, NY.
Georgios.Moutsanidis@stonybrook.edu Web page

Dr. Jason Wang
Distinguished Engineer
ANSYS LS-DYNA
Livermore, NY.
Jason.Wang@ansys.com

Dr. Edouard Yreux
Senior Principle R&D Engineer
ANSYS LS-DYNA
Livermore, NY.
Edouard.Yreux@ansys.com